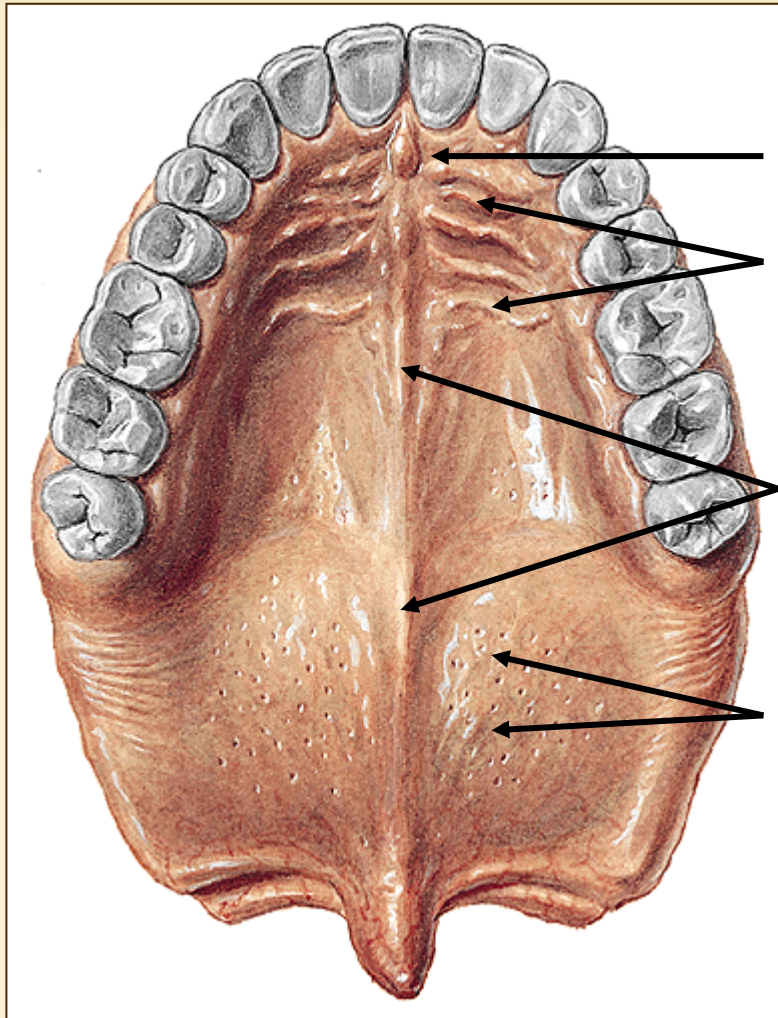


The roof of the oral cavity: **the palate**

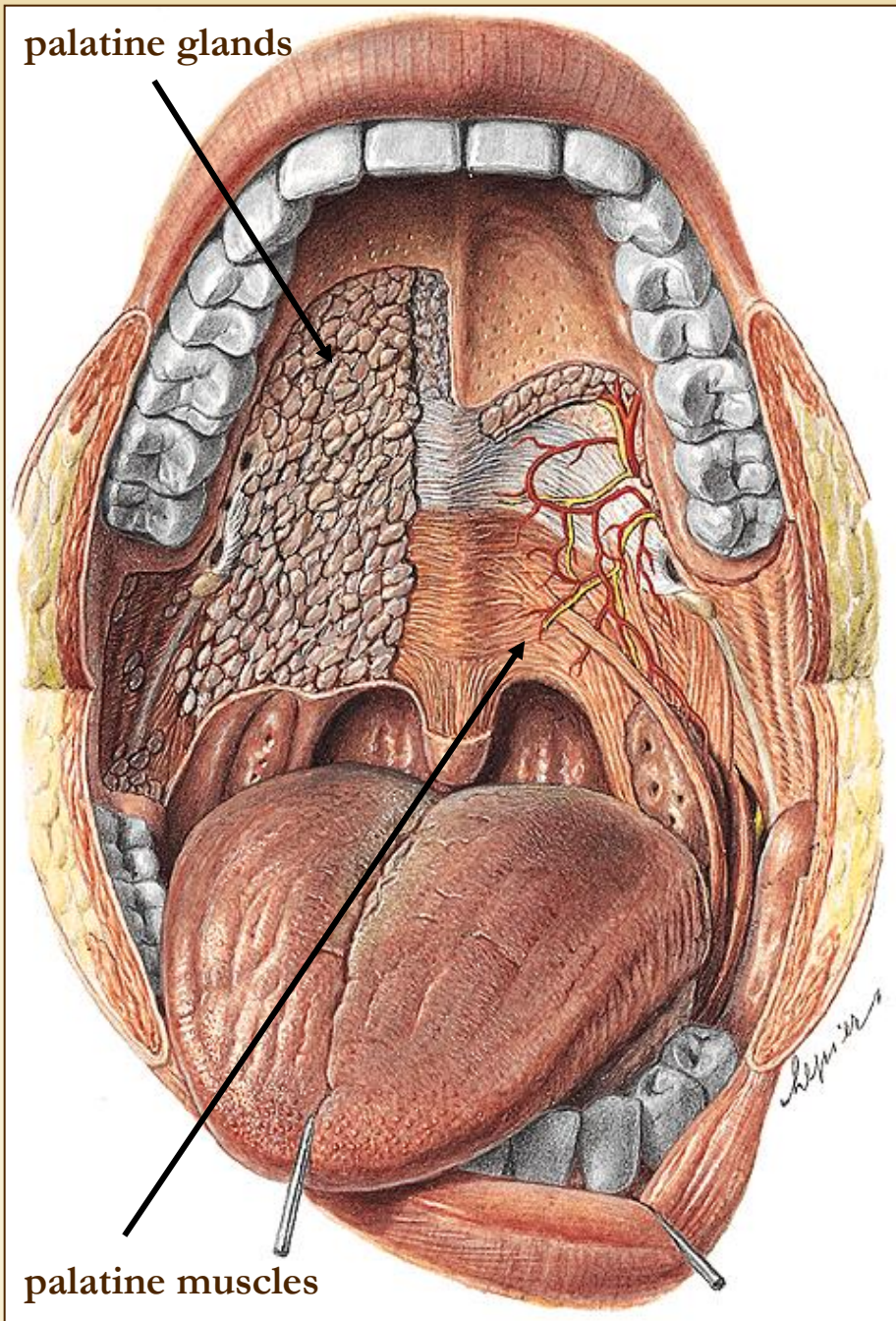


incisive papilla: pear-shaped swelling over the incisive foramen

palatine rugae: several (3-7) transverse folds of the mucosa of the hard palate (chopping and grinding of foods, phonation)

palatine raphe: longitudinal mucosal fold in the midline

palatine foveolae: openings of the excretory ducts of the small salivary glands of palate



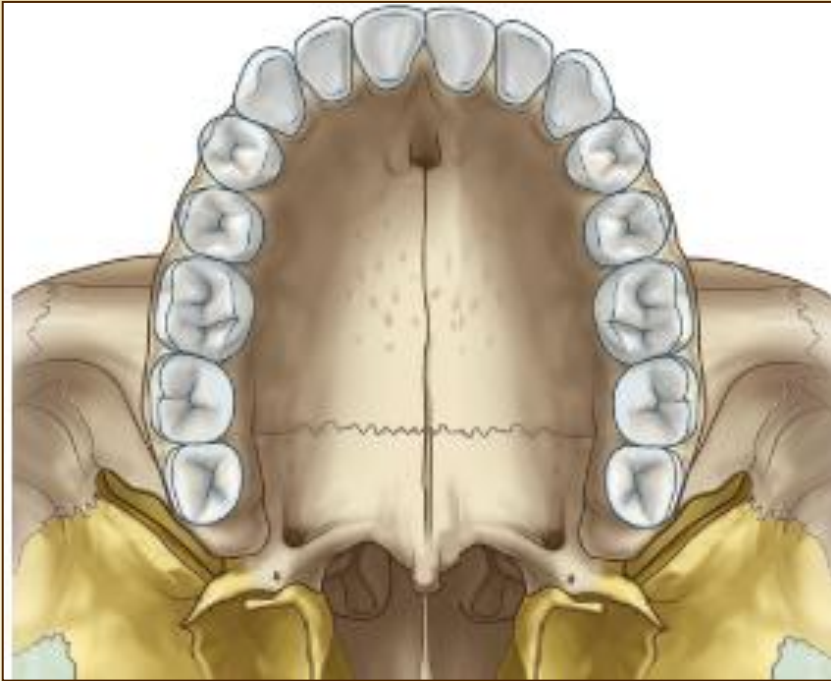
hard palate

Under the mucosa palatine glands are found, deep to them bony structures are seen.

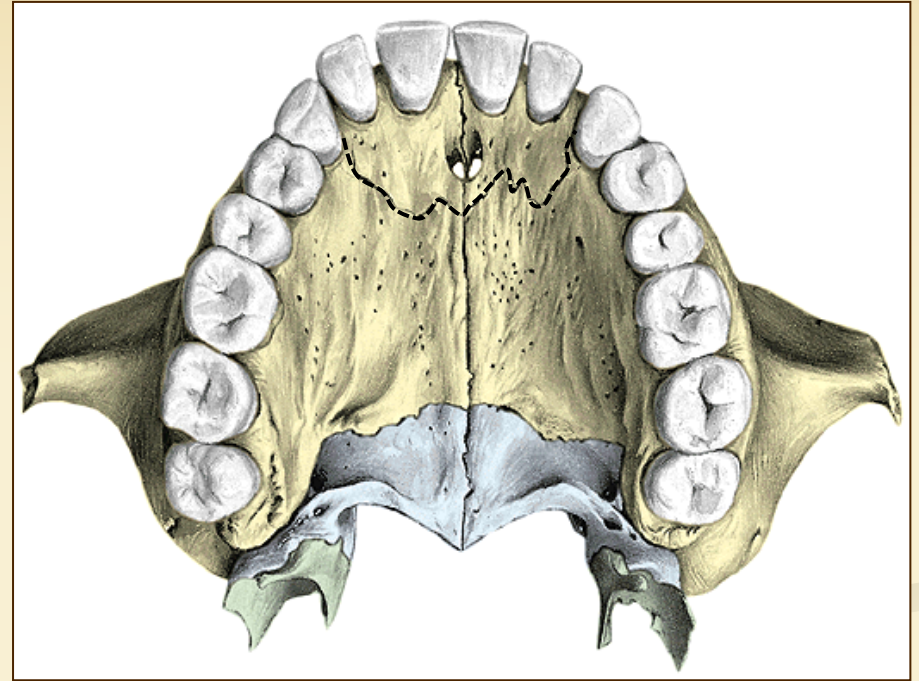
soft palate

Under the mucosa palatine glands are seen, deeper a fibromuscular structure (palatine muscles and palatine aponeurosis) is found.

Bony base of hard palate



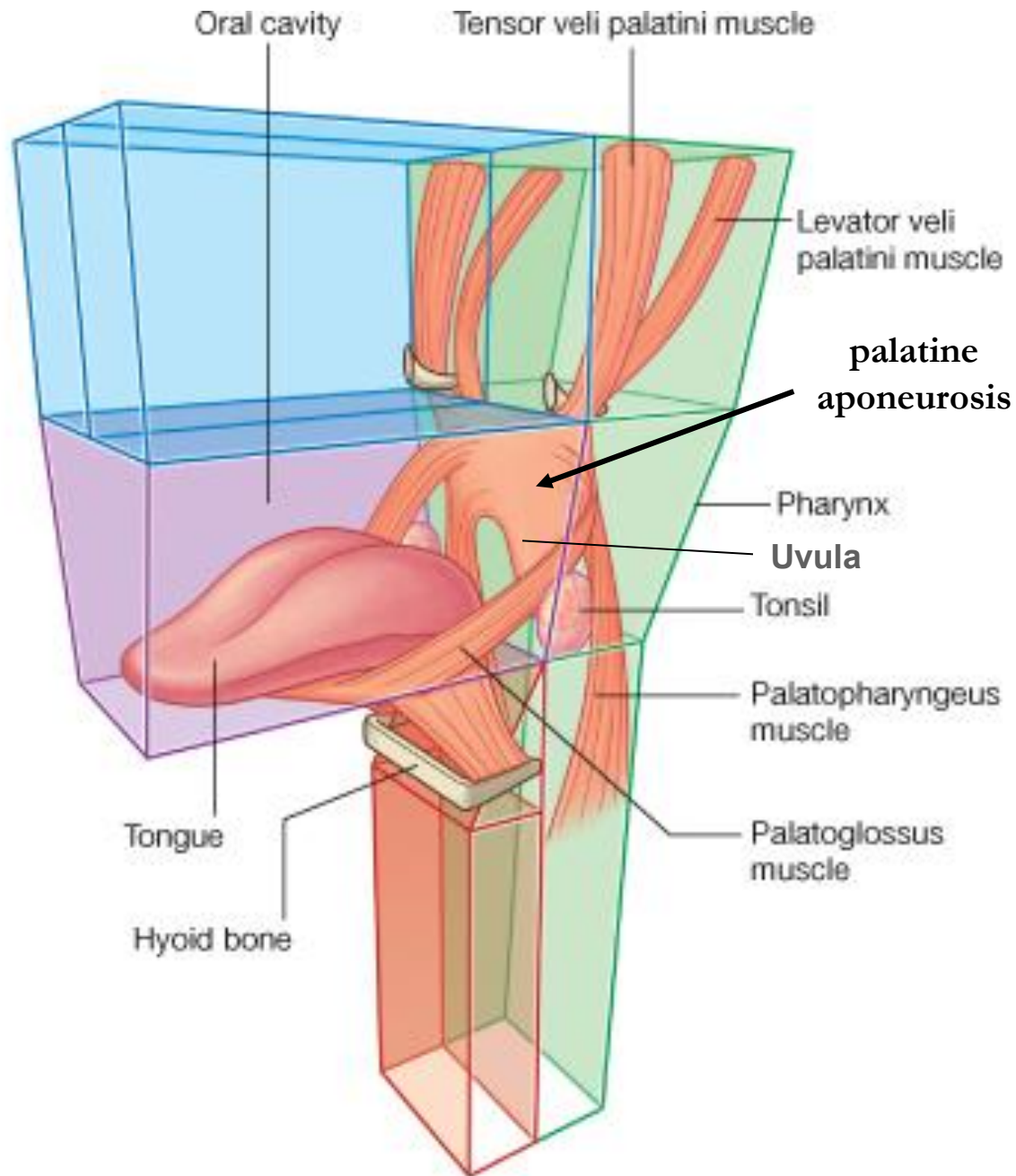
palatine proc. of maxillae
horizontal plate of palatine bone
median and transvers palatine sutures

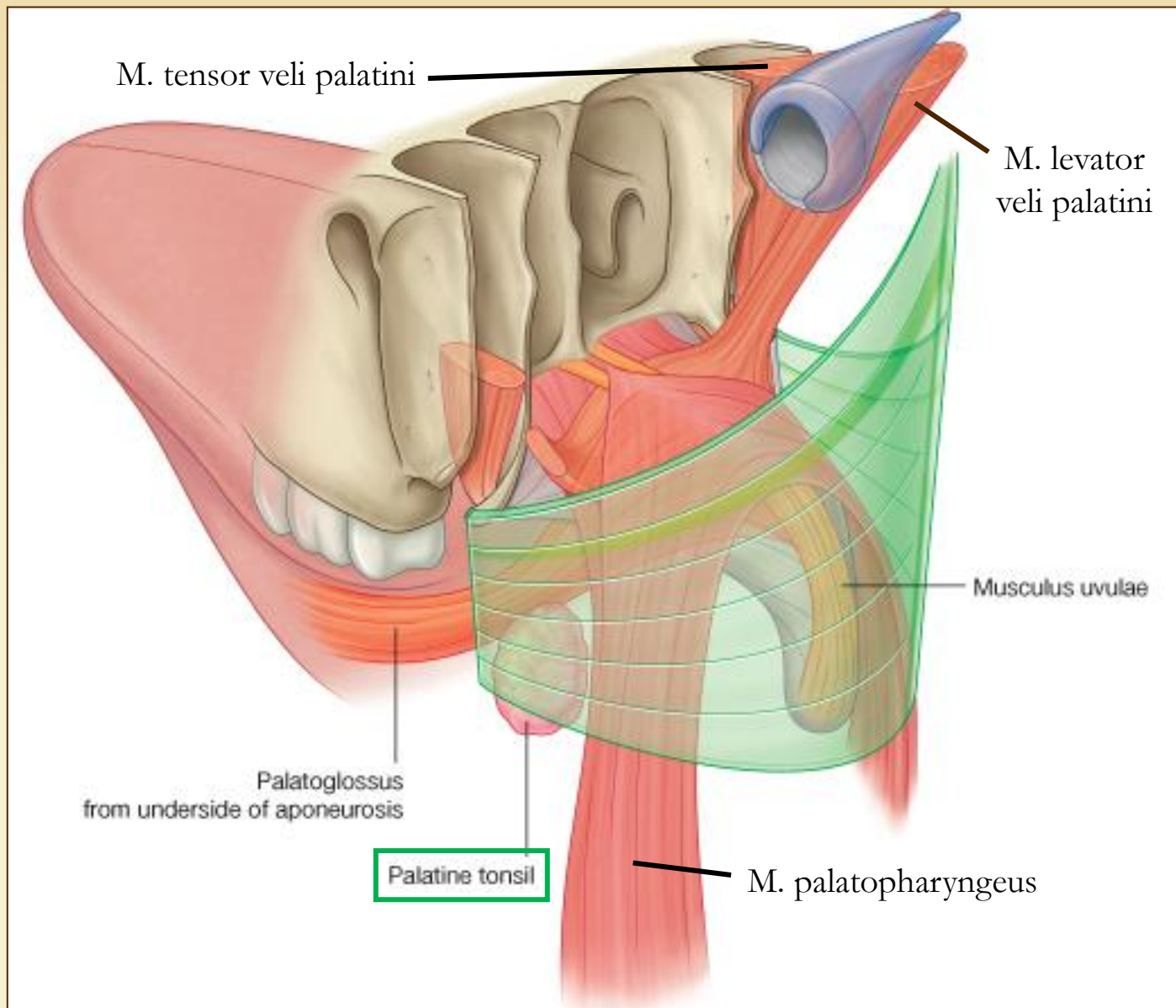


incisive canal
greater and lesser palatine foramina

Most of the mammals have obvious incisive suture and **incisive bone** (or **premaxilla**) but in human typically a complete fusion is seen.

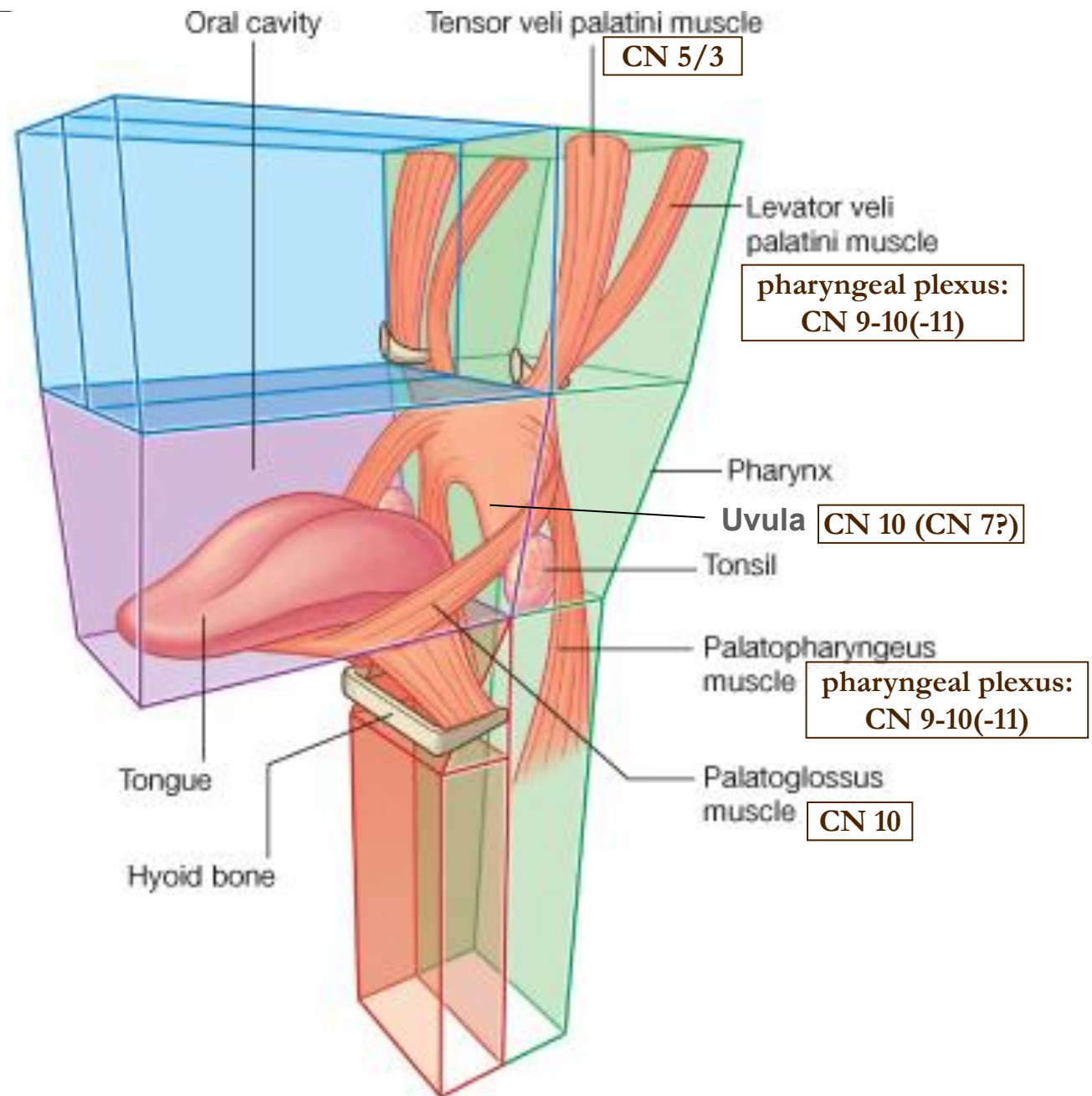
Muscles of the soft palate



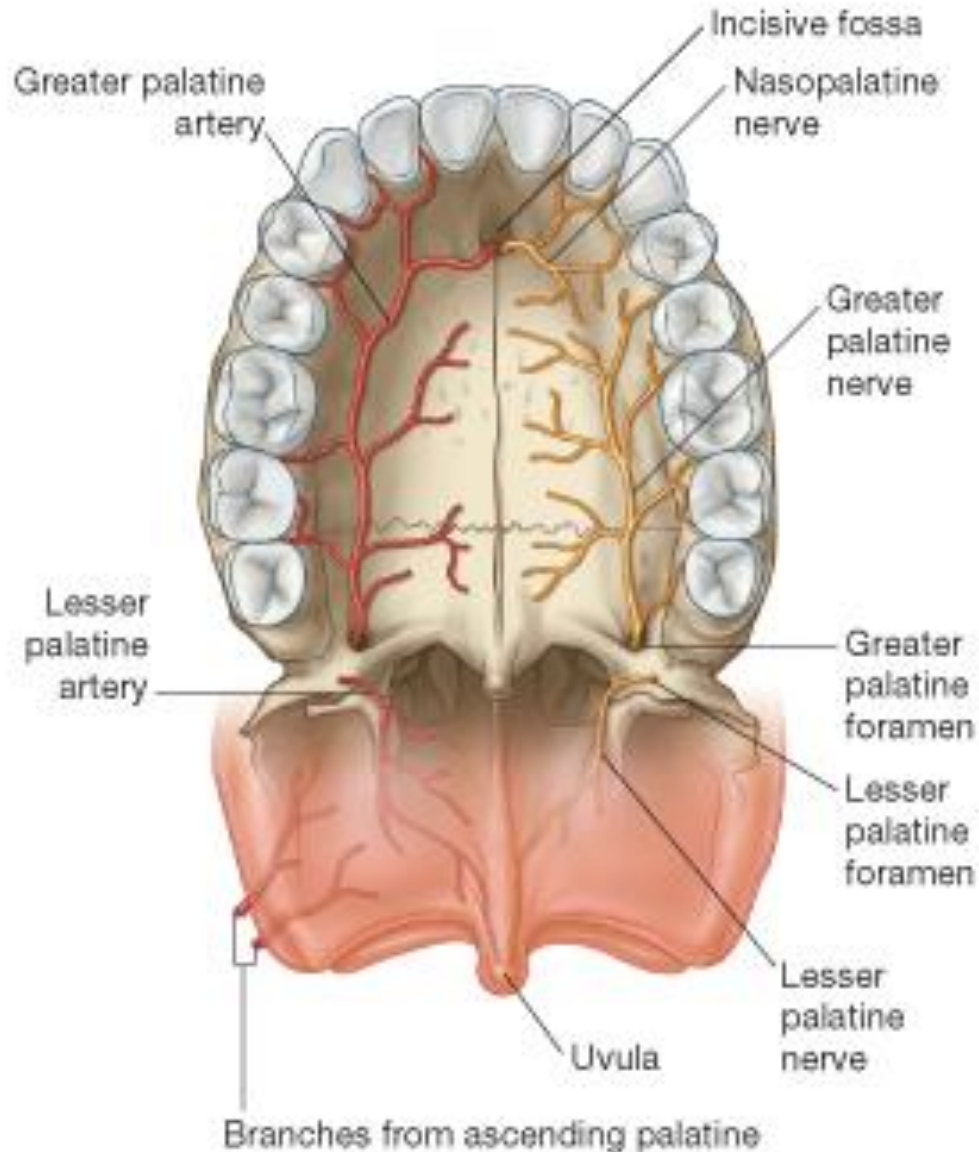


Tensor veli palatini and **levator veli palatini muscles** open the auditory (or Eustachian) tube at the moment of swallowing: this is essential for maintaining equilibrium of pressure on the two sides of the tympanic membrane (eardrum).

Innervation of palatine muscles



Innervation and blood supply of the palate



Descending palatine artery:

(labelled on the picture as greater palatine artery)
from sphenopalatine branch of maxillary artery coming from the pterygopalatine fossa

Incisive artery

terminal branch of nasopalatine artery arising from the sphenopalatine branch of maxillary artery

Ascending palatine artery

a direct branch from facial artery

Greater and lesser palatine nerves

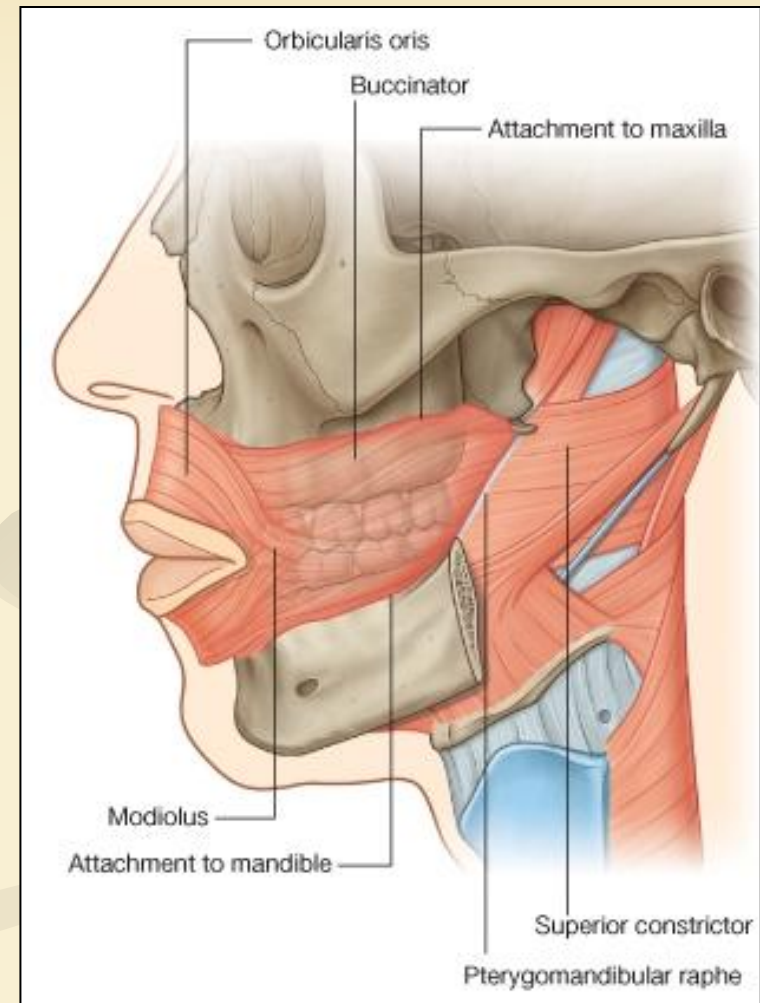
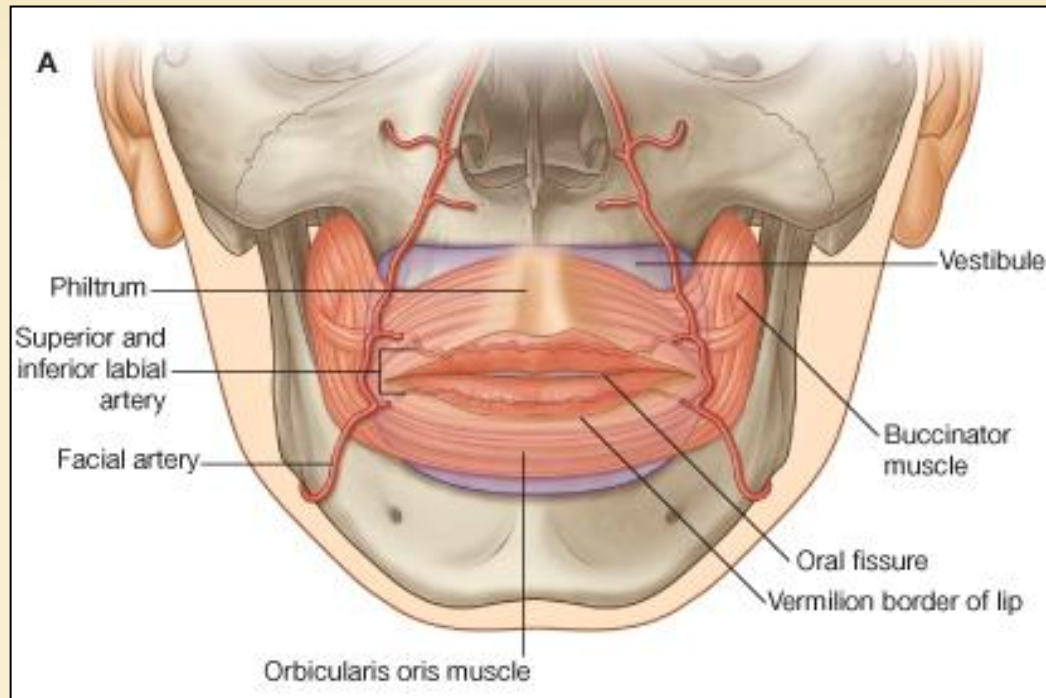
from ganglionic branches of maxillary nerve descending from the pterygopalatine fossa

Incisive nerve

terminal branch of nasopalatine nerve from ganglionic branches of the maxillary nerve

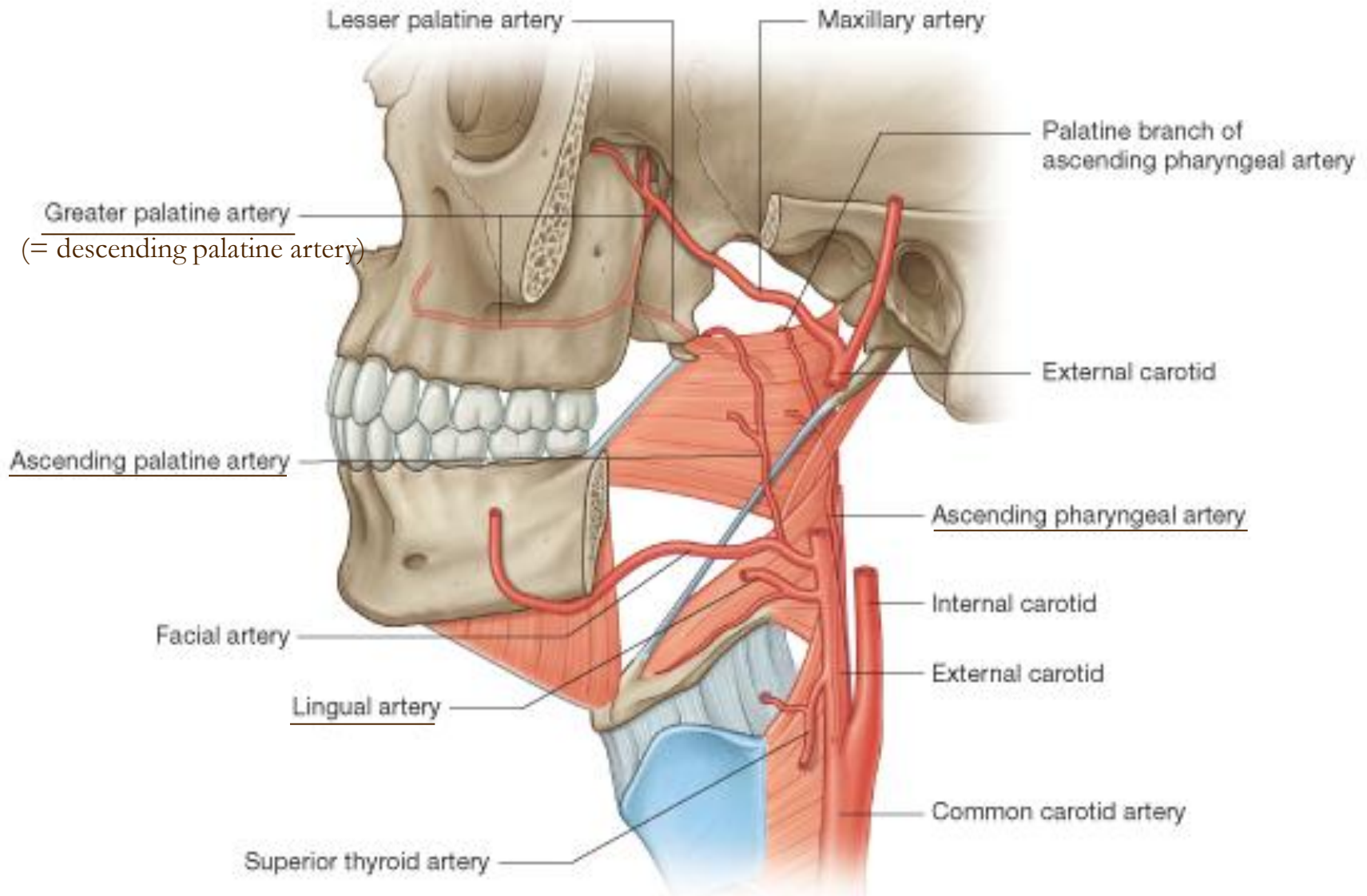
Note that arteries and nerves of the palate share the same name!

The anterolateral wall the oral cavity: **the lips and cheeks**



1. Skin
2. Orbicular oris m. / Buccinator muscle
3. Oral mucosa

Blood supply of the (palatine) tonsils



Production of saliva:

salivary glands

- Salivary glands produce 0.5 – 2.0 litres of saliva per day.
- Saliva keeps the oral mucosa moist, contains enzymes (amylase for digesting polysaccharides and lysozyme to destroy the wall of Gram-positive bacteria).
- Beside the three pairs of major salivary glands – the sublingual, submandibular and the parotid glands – minor glands also have to be mentioned.

Minor salivary glands

- Total number of minor salivary glands is estimated around 700-1000. Their diameter is 1-5mm and give approx. 5-8% of total saliva production.
- 3 sets of minor glands in the tongue: anterior or apical lingual gland (of Nuhn-Blandin) - mixed, posterior lingual glands (of Weber) - mucous, glands of the circumvallate and foliate papillae (of von Ebner) - serous.
- Palatine, buccal and labial glands.

Tongue

mucous

serous

Palatine Tonsil

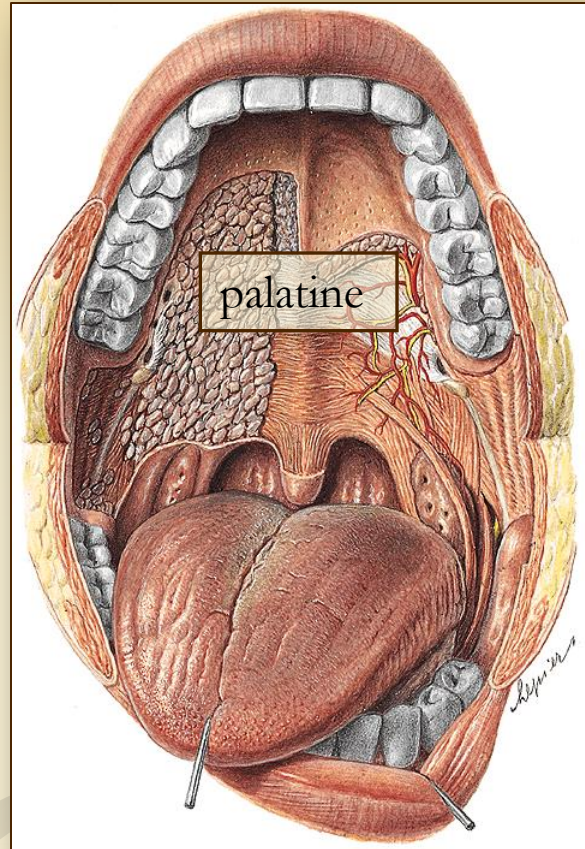
Lingual Tonsil

Circumvallate Papillae

Foliate Papillae

Filiform Papillae

Fungiform Papillae



Nuhn-Blandin's

von Ebner's

Weber's

